Zecheng Zhou



University of Macau · PhD Student
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Biophysical Modeling | Cardiac Electrophysiology | Epidemiology | Infectious Disease Modeling

Profile

PhD student at the University of Macau with experience in single-cell/single-nucleus transcriptomic analysis, cellular fluorescence imaging, and multiscale biophysical modeling. My cardiac virtual-cell research connects gene expression, cellular phenotypes, and dynamic signals with mechanistic models of ion-channel electrophysiology, calcium regulation, and sarcomere contraction.

Experienced in reanalyzing public snRNA-seq data using Seurat and Harmony, differential-expression and pseudobulk analysis, and quantitative fluorescence imaging with Fluo-4 and Di-8-ANEPPS. My work also involves cell culture, staining, electrical stimulation, functional assays, ROI quantification, and fluorescence time-series analysis.

My research focuses on AI-mechanistic integration, including Neural ODEs, neural-network/ODE hybrid models, multimodal data fusion, and data-driven parameter estimation for cell-state and perturbation-response prediction. Previously trained in Preventive Healthcare at Xiamen University, with experience in infectious disease modeling and chronic disease epidemiology.

Education

PhD Student — University of Macau, Macau SAR, China
Sep 2025 - Present

B.Sc. in Preventive Healthcare — Xiamen University, Fujian, China
Sep 2020 - Jun 2025 · GPA: 82.34 / 100 · CET-6: 504 · CET-4: 499

Technical Skills

- **Single-cell/single-nucleus omics:** Public snRNA-seq data reanalysis; Seurat normalization, Harmony batch correction, cell annotation, differential expression, and pseudobulk analysis.
- **Fluorescence imaging and experiments:** Fluo-4 calcium and Di-8-ANEPPS membrane-potential imaging; ROI quantification and fluorescence time-series analysis; cell culture, staining, electrical stimulation, and functional assays.
- **Multiscale mechanistic modeling:** ODEs, electrophysiology, subcellular calcium dynamics, sarcomere contraction, mechano-electric coupling, parameter optimization, and Sobol global sensitivity analysis.
- **AI-mechanistic integration:** Neural ODEs, neural-network/ODE hybrid models, multimodal data fusion, data-driven parameter estimation, and cell-state and perturbation-response prediction.
- **Programming and statistics:** R, Python, MATLAB, C/C++, SPSS, and Stata.
- **Additional analytical methods:** Logistic regression, chain binomial models, infectious disease modeling, PSM-DID, and basic metabolomics analysis.
- **Research tools:** EndNote, Microsoft Office, and GraphPad Prism.

Research Projects

Cardiac Virtual Cell Modeling: Multimodal Data, Biophysical Mechanisms, and AI

PhD Research Project · Aug 2025 - Present
University of Macau

Integrating public single-cell/single-nucleus transcriptomics, fluorescence imaging, and multiscale biophysical models to develop a virtual cell framework combining mechanistic and AI/data-driven approaches for predicting cardiac cell states and dynamic responses.

- **Single-cell/single-nucleus data analysis:** Reanalyzed public human atrial snRNA-seq data using Seurat normalization, Harmony batch correction, cell-subtype annotation, differential expression, and pseudobulk analysis to characterize cell states and disease-associated features.
- **Fluorescence imaging and wet-lab work:** Conducting cell culture, staining, electrical stimulation, and functional assays, including Fluo-4 calcium fluorescence and Di-8-ANEPPS membrane-potential imaging; analyzing ROI signals and fluorescence time series to quantify cellular dynamics.
- **Multiscale mechanistic modeling:** Coupled ODE systems for ion-channel electrophysiology, subcellular calcium regulation, and sarcomere contraction to model intracellular Ca^{2+} dynamics and active force generation.
- **Mechano-electric feedback and arrhythmia mechanisms:** Incorporated stretch-activated currents, mechanical feedback, and spatially heterogeneous calcium buffering; simulated calcium waves and delayed afterdepolarizations (DADs) across pacing frequencies.
- **Parameter analysis and multimodal calibration:** Performed Sobol global sensitivity analysis to quantify parameter effects on calcium transients and arrhythmia triggers; studying how transcriptomic features and imaging time series can constrain model parameters and support comparison with experimental observations.
- **AI-mechanistic integration:** Investigating Neural ODEs, neural-network/ODE hybrid models, and multimodal data fusion for data-driven parameter estimation, cell-state and perturbation-response prediction, and iterative model refinement.

Results:

- Presentation accepted at the 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS 2026), Munich, Germany, Jul 2026.

Image-Based Modeling of Liver Perfusion and Metabolism

PhD Research Project · Aug 2025 - Present
University of Macau

- Developing a digital liver perfusion model combining medical imaging, vascular flow modeling, and metabolic simulation.
- Reconstructed 3D liver geometry and vascular structures from CT imaging data.
- Built multi-scale perfusion models describing blood flow through hepatic vascular networks.
- Coupled perfusion with ODE-based metabolic models (oxygen, glucose, lactate transport).
- Investigated temperature-dependent changes in vascular resistance and perfusion dynamics relevant to ex-vivo organ perfusion.
- Developed computational pipelines for simulation of transport and metabolic activity in liver tissue.

Relationship between dietary sialic acid intake and serum HbA1c Project Leader · Jan 2024 - Present

Population-based cross-sectional study, Xiamen, China

- Conducted a cross-sectional analysis of the association between dietary sialic acid intake and HbA1c levels.
- Performed logistic regression and related statistical analyses using SPSS and R.
- Gained experience in nutrient analysis methods and English scientific writing.

Main results:

- Co-authored paper in *Scientific Reports* (JCR Q1): "Relationship between dietary sialic acid intake and serum HbA1c: a population-based study in Xiamen, China."
- Poster presented at the South China Public Health Conference & CHPAMS 2024 Wall Poster Showcase.

Sialic acid intervention, LDL sialylation, and prevention of atherosclerosis Project Leader · Jul 2023 - Present

- Used different substrate combinations to study trans-sialidase (TS) reaction characteristics.
- Explored modification of LDL by exogenous sialic acids and their potential as TS substrates.

Gains: basic laboratory operations and use of laboratory equipment. Results: team won second and third prizes in Xiamen University's Basic Medical Innovative Research and Experimental Design Forum.

Assessment of aggregated HFMD outbreaks in Fujian Province based on a chain binomial model

Project Leader · Jul 2022 - Jun 2023

- Established a chain binomial model based on the natural history of HFMD virus.
- Fitted the model to aggregated outbreak data in Fujian Province to estimate transmission parameters and predict cluster outbreaks.

Gains: strengthened skills in R, Python, infectious disease modeling, experimental planning, and project submission. Results: developed an "Online Geocoding Software Based on R Language" at ShenSoft.

Temporal shifts in 24 notifiable infectious diseases in China before and during COVID-19

Team Member · May 2023

- Crawled provincial-level surveillance data for 24 notifiable diseases.
- Contributed to modeling and prediction for selected diseases using R and Python.

Results:

- Co-authored paper in *Nature Communications* (IF 14.7, JCR Q1).
- Research reported on Xiamen University Science and Technology Department webpages and other media.

Enhancing probiotic effects of *Bifidobacterium bifidum* and preparation of engineered strains

Team Member · Dec 2023 - Present

- Performed metabolomics data analysis and screening of engineered strains.
- Learned project filing, patent drafting, and scientific writing.

Results: awarded Outstanding Project in the 8th Annual Innovation and Entrepreneurship Conference of Xiamen University.

Impact of long-term care insurance pilot on catastrophic medical expenditures in China

Leader · Apr 2024 - Present

- Evaluated the impact of long-term care insurance (LTCI) on catastrophic medical expenditures of urban and rural residents.
- Applied propensity score matching difference-in-differences (PSM-DID) models using Stata.

Results: manuscript under revision for *China Health Statistics* (Peking University Core / Xiamen University First Class Core Journal).

Academic Competitions & Awards

- **National:** College Students' Innovative Entrepreneurial Training Plan Program - National Project.
- **National:** Tenth National Forum for Undergraduate Innovation Research & Experimental Design Competition - Gold Award and Best Manner Award.
- **National:** 14th MathorCup Maths Applied Challenge 2024 - Second Prize (National), Provincial First Prize, Fourth Place in East China Competition.
- **School:** Xiamen University Outstanding Undergraduate Thesis - Second Prize.
- **School:** First and Second National Forum for Undergraduate Innovation Research & Experimental Design Competition, Xiamen University - Second and Third Prizes.
- **School:** 2022 Outstanding Achievements in Practical Teaching of Ideological and Political Theory Courses, Xiamen University - Second Prize.
- **Xiamen University Innovation & Entrepreneurship Conference:** Outstanding Project Award (Bifidobacterium project).

Selected Publications & Presentations

- **Zhou, Z.**, Zhang, W., Wang, X. *et al.* Relationship between dietary sialic acid intake and serum HbA1c: a population-based study in Xiamen, China. ***Scientific Reports***, 14, 23026 (2024).
- Li, K., Rui, J., Song, W., Luo, L., Zhao, Y., Qu, H., Liu, H., Wei, H., Zhang, R., Abudunaibi, B., Wang, Y., **Zhou, Z.**, Xiang, T., & Chen, T. Temporal shifts in 24 notifiable infectious diseases in China before and during the COVID-19 pandemic. ***Nature Communications***, 15, 3891 (2024).
- Li, K., Wu, J., Zhang, R., Xie, Y., **Zhou, Z.**, Yin, Q., Chen, Q., Rui, J., Guan, X., Zhao, Z., & Chen, T. Global, regional, and national burdens of pertussis among adults: a systematic analysis of age-specific trends using Global Burden of Diseases 2021 data. ***Infectious Diseases of Poverty***, 14, 85 (2025).
- **Zhou, Z.**, & Lei, C. Spatial remodeling of myolysis and troponin C buffering couples subcellular Ca^{2+} dynamics to contractile function in atrial fibrillation. Accepted conference presentation, **WCCM-ECCOMAS 2026**, MS341B: Imaging-Based Mathematical and Computational Modeling in Biomechanics II, Munich, Germany, Jul 2026.

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